JÉRÉMY FOELLER



Software Engineer | AI Systems & Architecture | School 42 Marseille, FR

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Profile: Backed by 10 years of rigorous project management and entrepreneurial execution, currently mastering advanced software engineering at School 42. Versatile engineer working across C and Python with a first-principles approach. Specializing in high-performance system architectures, AI-powered product engineering, and scalable, production-grade systems with end-to-end ownership.

LinkedIn : https://www.linkedin.com/in/jérémy-foeller-95207a70 | **GitHub:** https://github.com/jefo1383

Key Skills:

Languages & Systems:	C (Strict 42 standard, manual memory management, pointer arithmetic), Python 3 (Strict typing, OOP, Functional), Bash Scripting, Linux (Debian, Alpine).
Artificial Intelligence & Agents:	Autonomous LLM Agents, Advanced RAG Pipelines (Chunking, Dense Embeddings, Re-ranking), Tool/Function Calling, Constrained Decoding, Pydantic v2, SLM Integration.
DevOps, Network & Security:	Docker, Docker Compose, Nginx (TLS 1.3), TCP Sockets & Multi-client Servers, TCP/IP Routing, Encrypted LVM, UFW Firewall, SSH Hardening, Git & CI/CD workflows.
Architecture & Algorithms:	Graph Theory & Pathfinding Algorithms, Multithreaded Concurrency (POSIX Threads, Mutexes, Deadlock Prevention), Time & Space Complexity (Big-O), Agile Engineering.

Major Projects & Engineering Achievements:

Cyber AI Hackathon • Partnership with CMA CGM, Mistral AI & Tangram

- Autonomous AI Incident Agent: Designed and built an end-to-end autonomous agent for dynamic ingestion, classification, and real-time triage of SOC security incidents.
- Multimodal Ingestion & Remediation Pipelines: Engineered multimodal extraction (log parsing, screenshot OCR), threat categorization engines, automated remediation playbooks, and interactive real-time dashboards.

AI & Advanced Software Architectures • School 42

- **RAG Against The Machine & Agent Smith:** Developed robust, modular RAG architectures (semantic chunking, vector embeddings, re-ranking) and autonomous agent frameworks featuring Chain-of-Thought reasoning and self-healing validation loops.
- **Call Me Maybe (Tool & Function Calling Engine):** Implemented deterministic Function Calling via Constrained Decoding, achieving 100% structured JSON / Pydantic schema adherence using lightweight local SLMs (Qwen 0.6B).
- **Pac-Man (Object-Oriented Engine):** Engineered a full-featured game in Python using strict OOP principles, deterministic state management, autonomous ghost AI pathfinding (chase/scatter/flee behaviors), and dynamic config parsers.

Infrastructure, Network & Low Level • School 42

- **Inception & Born2beRoot:** Deployed a microservices infrastructure using isolated Docker containers (Nginx TLS 1.3, WordPress, MariaDB, Docker Secrets) on a hardened Debian server with encrypted LVM partitions and strict security policies.
- **The Answer Protocol:** Built a high-concurrency TCP multiplayer server in C implementing a custom RFC-compliant network protocol with strict ABNF grammar parsing, asynchronous socket polling, and robust state serialization.
- **Fly-in & Codexion:** Designed capacity-constrained network routing algorithms using spatio-temporal BFS in Python; engineered multithreaded simulation kernels in C using POSIX threads, mutexes, and deadlock-free FIFO/EDF scheduling.

Professional Experience (10 years in Project Management):

2020 - 2025 | JERA Design:	*Interior Architect • Entrepreneur* – Directed end-to-end technical scoping, structural feasibility studies, budget allocation, and multi-stakeholder project delivery with absolute accountability.
2020 - 2021 | MMI Déco:	*Trainer* – Delivered pedagogical training and structured methodologies for national RNCP state
2013 - 2025 | Retail Design:	*Design Consultant & Project Lead (BoConcept, Mobalpa, IKEA) – Managed complex client requirements, high-value technical sales, and custom spatial implementations.*
2008 - 2010 | Gendarmerie:	*Police Officer – Developed deep procedural discipline, high-stress resilience, operational rigor, and selfless team spirit.*

Education & Personal Profile:

School 42 Marseille (2025 - Present): Common Core • **ESDAC (2014):** Bachelor's degree in Interior Design • **Bac S (2006):** High School Diploma in Sciences

Languages: French (Native), English (B2-C1) • **Interests:** Karate black belt, Tennis, Design, Tech.